ERASMUS+ INTERNATIONAL PHD SUMMER SCHOOL 2025
Mathematics and Machine Learning for image analysis
Lecture 3 - Learning Optimal Discretizations

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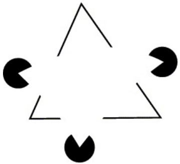
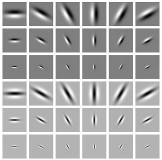
Edges

- **Edges** are among the most important features in images
- Image understanding relies on abstract discontinuity information
- Most successful image descriptors are based on intensity gradients
- First layers in deep convolutional networks represent edge detectors

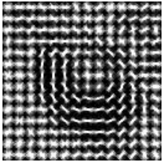
(a)



(b)



(c)



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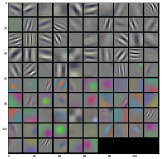
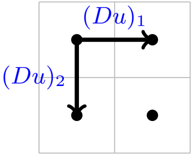
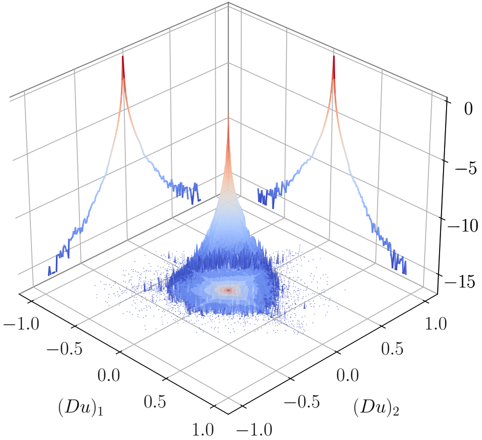


(e)

Edge statistics of natural images

- Randomly extracted $15M$ image patches of size 2×2 from a natural image data set.
- Compute finite differences in horizontal and vertical direction.
- Yields a heavy tailed distribution $\rightsquigarrow$ most gradients are zero $\rightsquigarrow$ sparse gradients.



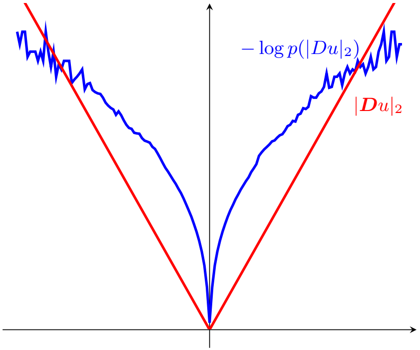


The total variation

- The log-statistics of Du looks like an upside-down ice-cream cone.
- A simple fit to the negative log-statistics is given by the ℓ_2 norm, leading to the total variation:

$$\mathrm{TV}(u) = \|Du\|_{2,1} = \sum_{i,j} \sqrt{(u_{i+1,j} - u_{i,j})^2 + (u_{i,j+1} - u_{i,j})^2}.$$

- Has been introduced in [Rudin, Osher, Fatemi '92], [Chambolle, Lions '97], ...



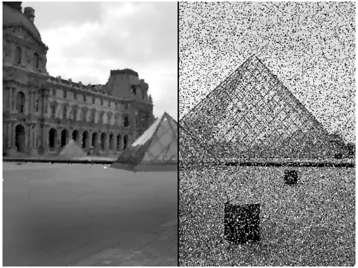
The ROF model

- The discrete ROF model [Rudin, Osher, Fatemi '92] is defined as the following minimization problem

$$\min_u \lambda \|Du\|_{2,1} + \frac{1}{2}\|u-g\|^2, \ \lambda > 0$$

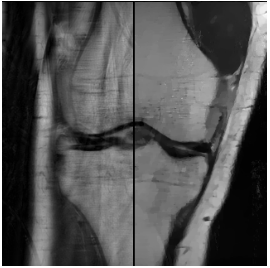
- Defines "the" prototypical variational model in mathematical image processing.
- Gives a good tradeoff between simplicity of the model and denoising quality.
- Allows for discontinuities (edges) in the image.
- It is a convex lower-semicontinuous function.
- It also has a nice geometric interpretation $\rightsquigarrow$ minimal surfaces.

Advanced applications

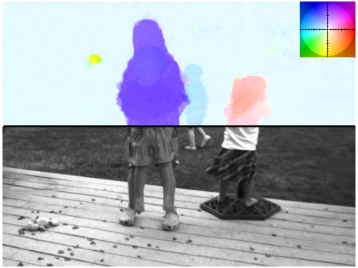


(a) Denoising

(b) Deblurring



(c) MRI



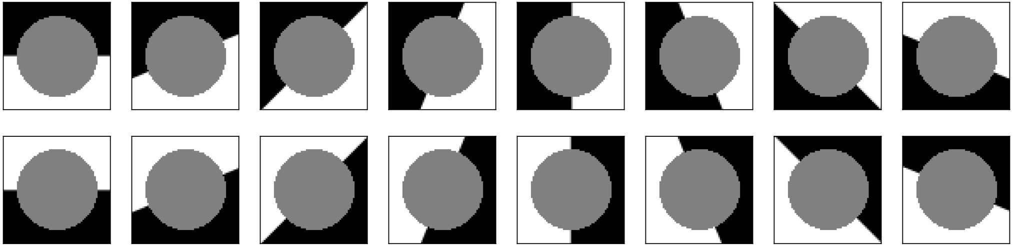
(d) Motion

(e) Stereo

(f) Segmentation

Image inpainting

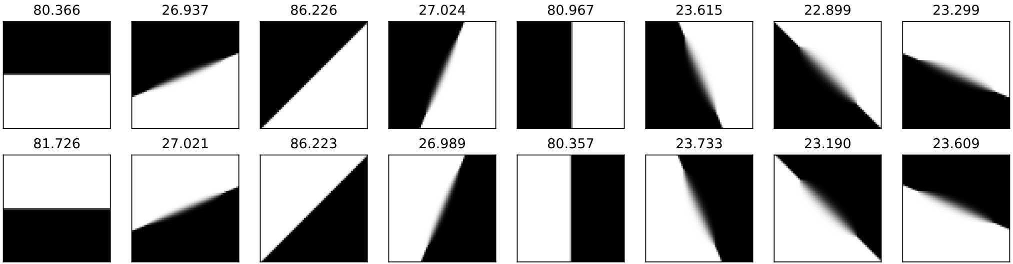
- For most practical problems, the standard discrete total variation gives sufficiently good results.
- However, on free discontinuity problems such as image inpainting, the standard discretization yields strong artifacts.



Inpainting of straight discontinuities

Image inpainting

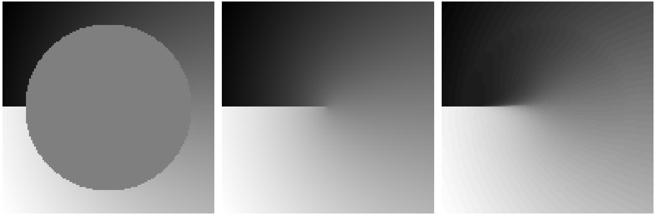
- For most practical problems, the standard discrete total variation gives sufficiently good results.
- However, on free discontinuity problems such as image inpainting, the standard discretization yields strong artifacts.



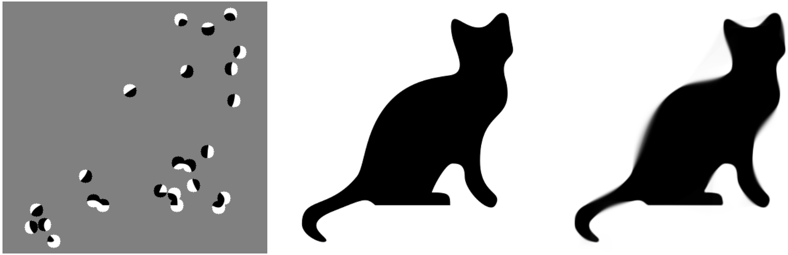
Inpainting of straight discontinuities

Advanced free discontinuities problems

Convexification of the Mumford-Shah functional [P., Cremers, Bischof, Chambolle '09]:



Convexification of Euler's elastica [Chambolle, P. '19]



(data from J. Weickert)

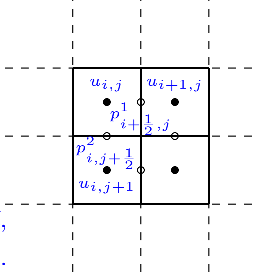
Here, the discretization can make a difference between "working" and "not working".

Related work

Finding a good general discretization of the total variation is far from being trivial and hence many approaches have been proposed:

- Non-standard finite differences for anisotropic diffusion [Weickert, Welk, Wichert '13]
- Graph-based / MRFs / crystalline energies [Boykov, Kolmogorov '03], [Chambolle '05]
- Upwind discretization [Chambolle, Levine, Lucier '11]
- Shannon TV [Abergel, Moisan '17]
- Conforming P1 finite elements [Bartels '12]
- Non-conforming P1 (Crouzeix-Raviart) finite elements [Chambolle, P. 18]
- Duality based discretization using $H(\mathrm{div})$-conforming Raviart-Thomas (RT0) vector fields [Herrmann, Herzog, Schmidt, Vidal, Wachsmuth '18], [Caillaud, Chambolle '20]
- Approximate Raviart-Thomas [Hintermüller, Rautenberg, Hahn '14], **[Condat '17]**

General setting



- We introduce the finite differences operator $Du = (D^1 u, D^2 u)$ with

$$\begin{cases} (D^1 u)_{i+\frac{1}{2},j} = u_{i+1,j} - u_{i,j} & i = 1, \dots, M-1, \ j = 1, \dots, N, \\ (D^2 u)_{i,j+\frac{1}{2}} = u_{i,j+1} - u_{i,j} & i = 1, \dots, M, \ j = 1, \dots, N-1. \end{cases}$$

- The total variation is defined via duality as

$$TV(u) := \sup \left\{ \langle \boldsymbol{p}, Du \rangle_Y : \|\boldsymbol{F}\boldsymbol{p}\|_{Z}{}^* \leq 1 \right\}$$

  where $\boldsymbol{p} = (p^1, p^2)$ are the dual variables and $\boldsymbol{F} = (\boldsymbol{F}^1, ..., \boldsymbol{F}^L)$ are convolutional interpolation kernels defined as

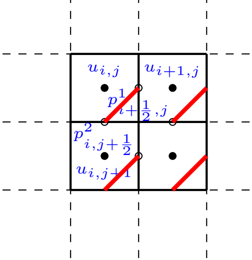
$$(\boldsymbol{F}^l \boldsymbol{p})_{i,j} = \begin{pmatrix} (F^{l,1} p^1)_{i,j} \\ (F^{l,2} p^2)_{i,j} \end{pmatrix} = \begin{pmatrix} \sum_{m,n=-\nu}^{\nu} \xi^l_{m,n} p^1_{i+\frac{1}{2}-m,j-n} \\ \sum_{m,n=-\nu}^{\nu} \eta^l_{m,n} p^2_{i-m,j+\frac{1}{2}-n} \end{pmatrix}$$

- The primal form has the structure of a sparse coding problem

$$TV(u) = \min_{\boldsymbol{q} :\, \boldsymbol{F}^* \boldsymbol{q} = Du} \|\boldsymbol{q}\|_Z ,$$

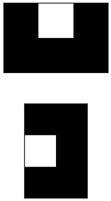
  where $\boldsymbol{F}^*$ can be interpreted as a convolutional dictionary.

Example: Forward differences



- Interpolation kernels (Nearest neighbor interpolation):

$$(Fp)_{i,j} = \begin{pmatrix} p^1_{i+\frac{1}{2},j} \\ p^2_{i,j+\frac{1}{2}} \end{pmatrix}.$$

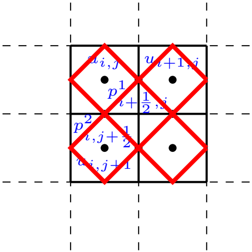


Interpolation kernels F

- The Z-norm is given by

$$\|\boldsymbol{z}\|_Z = \sum_{i,j} \sqrt{(z^1_{i+\frac{1}{2},j})^2 + (z^2_{i,j+\frac{1}{2}})^2}$$

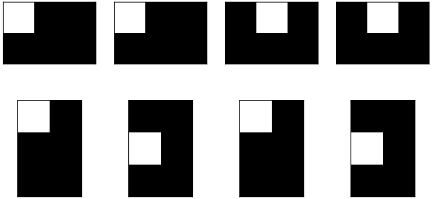
Example: Raviart-Thomas



- Interpolation kernels (Nearest neighbor interpolation):

$$(F^1 p)_{i-\frac{1}{2},j-\frac{1}{2}} = \begin{pmatrix} p^1_{i-\frac{1}{2},j} \\ p^2_{i,j-\frac{1}{2}} \end{pmatrix}, (F^2 p)_{i-\frac{1}{2},j+\frac{1}{2}} = \begin{pmatrix} p^1_{i-\frac{1}{2},j} \\ p^2_{i,j+\frac{1}{2}} \end{pmatrix},$$

$$(F^3 p)_{i+\frac{1}{2},j-\frac{1}{2}} = \begin{pmatrix} p^1_{i+\frac{1}{2},j} \\ p^2_{i,j-\frac{1}{2}} \end{pmatrix}, \quad (F^4 p)_{i+\frac{1}{2},j+\frac{1}{2}} = \begin{pmatrix} p^1_{i+\frac{1}{2},j} \\ p^2_{i,j+\frac{1}{2}} \end{pmatrix}.$$

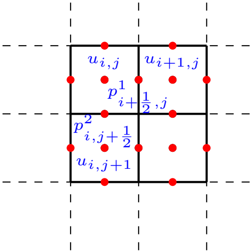


Interpolation kernels F

- The Z-norm is given by

$$\left\|(\boldsymbol{z}^1, \boldsymbol{z}^2, \boldsymbol{z}^3, \boldsymbol{z}^4)\right\|_Z := \sum_{i,j} |\boldsymbol{z}^1_{i-\frac{1}{2},j-\frac{1}{2}}|_2 + |\boldsymbol{z}^2_{i-\frac{1}{2},j+\frac{1}{2}}|_2 + |\boldsymbol{z}^3_{i+\frac{1}{2},j-\frac{1}{2}}|_2 + |\boldsymbol{z}^4_{i+\frac{1}{2},j+\frac{1}{2}}|_2$$

Example: Condat's discretization



- Interpolation kernels (bilinear interpolation):

$$(F^1 p)_{i,j} = \begin{pmatrix} \frac{p^1_{i-\frac{1}{2},j} + p^1_{i+\frac{1}{2},j}}{2} \\ \frac{p^2_{i,j-\frac{1}{2}} + p^2_{i,j+\frac{1}{2}}}{2} \end{pmatrix},$$

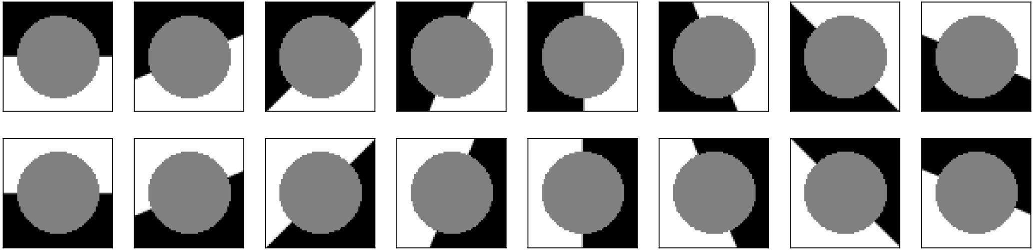
$$(F^2 p)_{i+\frac{1}{2},j} = \begin{pmatrix} p^1_{i+\frac{1}{2},j} \\ \frac{p^2_{i,j-\frac{1}{2}} + p^2_{i,j+\frac{1}{2}} + p^2_{i+1,j-\frac{1}{2}} + p^2_{i+1,j+\frac{1}{2}}}{4} \end{pmatrix}, \quad (F^3 p)_{i,j+\frac{1}{2}} = \begin{pmatrix} \frac{p^1_{i-\frac{1}{2},j} + p^1_{i+\frac{1}{2},j} + p^1_{i-\frac{1}{2},j+1} + p^1_{i+\frac{1}{2},j+1}}{4} \\ p^2_{i,j+\frac{1}{2}} \end{pmatrix}$$

Interpolation kernels F

- The Z-norm is given by

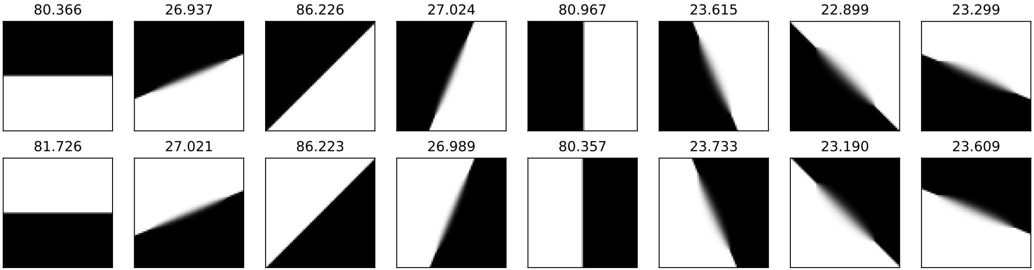
$$\|(\boldsymbol{z}^1, \boldsymbol{z}^2, \boldsymbol{z}^3)\|_Z := \sum_{i,j} |\boldsymbol{z}^1_{i,j}|_2 + |\boldsymbol{z}^2_{i+\frac{1}{2},j}|_2 + |\boldsymbol{z}^3_{i,j+\frac{1}{2}}|_2$$

Comparison



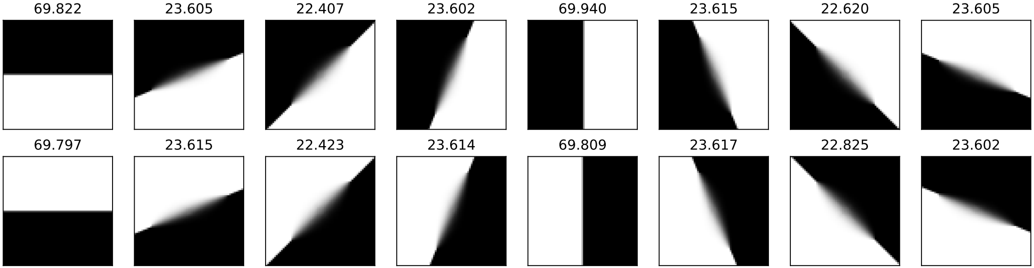
Input

Comparison



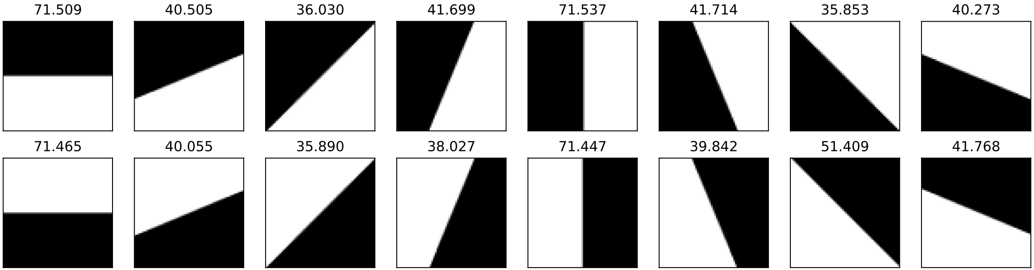
Forward differences

Comparison



Raviart-Thomas

Comparison



Condat

Consistency result

- We define a family of discrete total variations for pixels of size $\varepsilon \times \varepsilon$:

$$TV_\varepsilon(u) = \min\left\{\varepsilon^2 \|\boldsymbol{q}\|_{Z_\varepsilon} : \boldsymbol{F}_\varepsilon^* \boldsymbol{q} = D_\varepsilon u\right\} = \sup\left\{\varepsilon^2 \langle \boldsymbol{p}, D_\varepsilon u\rangle_{Y_\varepsilon} : \|\boldsymbol{F}_\varepsilon \boldsymbol{p}\|_Z^* \leq 1\right\}$$

Theorem

Assume the supports and the weights of the convolutions defining $\boldsymbol{F}_\varepsilon$ are uniformly bounded that is

$$\sum_{m,n} \xi_{m,n}^l = \sum_{m,n} \eta_{m,n}^l = 1 \iff F^{l,1}, F^{l,2} \in C_{\Sigma=1}.$$

Then TV_ε Γ-converges to

$$TV(u) := \begin{cases} |Du|(\Omega) & \textit{if } u \in BV(\Omega), \\ +\infty & \textit{else.} \end{cases}$$

As long as the filter coefficients sum up to one and are uniformly bounded, we are having a consistent discretization of the total variation $\rightsquigarrow$ learning.

Class of total variation minimization problems

- We consider the following class of total variation minimization problems

$$\min_{Du=\boldsymbol{F}^*\boldsymbol{q}} \lambda \|\boldsymbol{q}\|_Z + G(u,g),$$

  with a saddle-point formulation

$$\min_{u,\boldsymbol{q}} \max_{\boldsymbol{p}} \langle Du - \boldsymbol{F}^*\boldsymbol{q}, \boldsymbol{p}\rangle + \lambda \|\boldsymbol{q}\|_Z + G(u,g)$$

- Can be applied to a large class of inverse problems in imaging such as denoising, inpainting, segmentation, ...
- We need access to the proximal maps of $\|\cdot\|_Z$ and $G(\cdot,g)$.

Supervised learning

- Assume we have given training data (g_s, t_s), $s = 1, ..., S$.
- We consider the following bilevel optimization problem:

$$\min_{\boldsymbol{F}} \mathcal{L}(\boldsymbol{F}) + \mathcal{R}(\boldsymbol{F}),$$
$$u_s^* \in \arg\min_{u,\boldsymbol{q}} \max_{\boldsymbol{p}} \langle Du - \boldsymbol{F}^* \boldsymbol{q}, \boldsymbol{p} \rangle + \lambda \|\boldsymbol{q}\|_Z + G(u, g_s), \quad s = 1, \dots, S$$

- $\mathcal{L}(\boldsymbol{F})$ is a convex and differentiable loss function

$$\mathcal{L}(\boldsymbol{F}) = \frac{1}{MNS} \sum_{s=1}^{S} \ell(u_s^*(\boldsymbol{F}), t_s),$$

that measures the error between the targets t_s and the solutions u_s^*, here $\ell(u, t) = \frac{1}{2} \|u - t\|_2^2$.
- $R(\boldsymbol{F})$ can be used to impose the constraints on the filters $\boldsymbol{F}$.

$$\mathcal{R}(\boldsymbol{F}) = \delta_{(C_{\Sigma=1})^{L,2}}(\boldsymbol{F}) = \sum_{l=1}^{L} \delta_{C_{\Sigma=1}}(F^{l,1}) + \delta_{C_{\Sigma=1}}(F^{l,2})$$

- For gradient-based learning, we need to compute the derivatives of the loss function with respect to the linear operator $\boldsymbol{F}$.

Interlude: Derivatives of saddle-points

- We consider the following class of saddle-point problems

$$\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} \langle Kx,\, y\rangle + g(x) - f^*(y),$$

  with corresponding primal and dual problems

$$\min_{x\in\mathcal{X}} f(Kx) + g(x) \Longleftrightarrow \max_{y\in\mathcal{Y}} -f^*(y) - g^*(-K^*y)$$

- We assume that the problem has, for a given linear operator K, a unique saddle point $(\hat{x}, \hat{y})$ characterized by

$$\begin{cases} K\hat{x} - \partial f^*(\hat{y}) \ni 0 \\ K^*\hat{y} + \partial g(\hat{x}) \ni 0 \end{cases}$$

- Then we consider that we have given a convex loss function

$$\mathcal{L}(K) = \ell(\hat{x}(K), \hat{y}(K)).$$

- We are interested in the gradient of the loss with respect to the linear operator K.
- We will derive a formula based on a standard sensitivity analysis.
- Denote by $\hat{x}_s = \hat{x} + s\xi_s$, and $\hat{y}_s = \hat{y} + s\eta_s$ the solution of the saddle-point problem perturbed by a small variation $K + sL$, $|s| \ll 1$ of the linear operator.

- Substituting $(\hat{x}_s, \hat{y}_s)$ into the optimality system yields

$$\begin{cases} K\hat{x} + s(K\xi_s + L\hat{x}_s) - [\partial f^*(\hat{y}) + (\int_0^s D^2 f^*(\hat{y} + t\eta_s)\mathrm{d}t)\eta_s] = 0, \\ K^*\hat{y} + s(K^*\eta_s + L^*\hat{y}_s) + [\partial g(\hat{x}) + (\int_0^s D^2 g(\hat{x} + t\xi_s)\mathrm{d}t)\xi_s] = 0. \end{cases}$$

- Again making use of the optimality condition and dividing by s gives

$$\begin{cases} K\xi_s + L\hat{x}_s - \left(\frac{1}{s}\int_0^s D^2 f^*(\hat{y} + t\eta_s)\mathrm{d}t\right)\eta_s = 0, \\ K^*\eta_s + L^*\hat{y}_s + \left(\frac{1}{s}\int_0^s D^2 g(\hat{x} + t\xi_s)\mathrm{d}t\right)\xi_s = 0. \end{cases}$$

- Passing to the limit $s \to 0$, one obtains the linear system in (ξ, η)

$$\begin{cases} K\xi + L\hat{x} - D^2 f^*(\hat{y})\eta = 0, \\ K^*\eta + L^*\hat{y} + D^2 g(\hat{x})\xi = 0. \end{cases} \iff \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} D^2 g(\hat{x}) & K^* \\ -K & D^2 f^*(\hat{y}) \end{pmatrix}^{-1} \begin{pmatrix} -L^*\hat{y} \\ L\hat{x} \end{pmatrix}$$

- The directional derivative is then given by

$$\begin{aligned} \mathcal{L}'(K;L) &= \left\langle \nabla\ell(\hat{x},\hat{y}), \begin{pmatrix} \xi \\ \eta \end{pmatrix} \right\rangle = \left\langle \nabla\ell(\hat{x},\hat{y}), \begin{pmatrix} D^2 g(\hat{x}) & K^* \\ -K & D^2 f^*(\hat{y}) \end{pmatrix}^{-1} \begin{pmatrix} -L^*\hat{y} \\ L\hat{x} \end{pmatrix} \right\rangle \\ &= \left\langle \underbrace{\begin{pmatrix} D^2 g(\hat{x}) & K^* \\ -K & D^2 f^*(\hat{y}) \end{pmatrix}^{-1} \nabla\ell(\hat{x},\hat{y})}_{=\begin{pmatrix} -\hat{X} \\ \hat{Y} \end{pmatrix}}, \begin{pmatrix} -L^*\hat{y} \\ L\hat{x} \end{pmatrix} \right\rangle = \left\langle \hat{X}, L^*\hat{y} \right\rangle + \left\langle \hat{Y}, L\hat{x} \right\rangle, \end{aligned}$$

where $\hat{X}$ and $\hat{Y}$ being the adjoint variables.

- Interestingly, the adjoint variables are themselves solutions of the quadratic saddle-point problem

$$\min_X \max_Y \langle KX, Y\rangle + \frac{1}{2}\langle D^2 g(\hat{x})X, X\rangle - \frac{1}{2}\langle D^2 f^*(\hat{y})Y, Y\rangle + \left\langle \nabla \ell(\hat{x}, \hat{y}), \begin{pmatrix} X \\ Y \end{pmatrix} \right\rangle$$

- This brings up the idea of running in parallel, a primary primal-dual algorithm solving the lower-level problem and a secondary primal-dual algorithm that solves the adjoint saddle-point problem.
- Such algorithmic scheme is denoted in the AD literature as "piggyback" algorithm [Griewank, Faure '03].
- Note that the secondary primal-dual algorithm depends on the solution of the primary primal-dual algorithm and hence must be analyzed as an algorithm with (summable) errors.
- The final gradient is then given by

$$\mathcal{L}'(K; L) = \left\langle \hat{X}, L^* \hat{y} \right\rangle + \left\langle \hat{Y}, L\hat{x} \right\rangle \iff \nabla \mathcal{L}(K) = \hat{X} \otimes \hat{y} + \hat{x} \otimes \hat{Y},$$

which we usually compute via automatic differentiation of the scalar products, in order to respect the structure and boundary conditions of the linear operator K (which can be complicated).

Piggyback primal-dual algorithm

- The primary primal-dual algorithm is given by

$$\begin{cases} x^{k+1} = (I + \tau \nabla g))^{-1}(x^k - \tau K^* y^k) \\ \bar{x}^{k+1} = x^{k+1} + \theta(x^{k+1} - x^k) \\ y^{k+1} = (I + \sigma \nabla f^*)^{-1}(y^k + \sigma K \bar{x}^{k+1}). \end{cases}$$

- The secondary primal-dual algorithm is given by

$$\begin{cases} X^{k+1} = \nabla \operatorname{prox}_{\tau g}(x^k - \tau K^* y^k) \cdot (X^k - \tau(K^* Y^k + \nabla_x \ell(x^k, y^k))) \\ \bar{X}^{k+1} = X^{k+1} + \theta(X^{k+1} - X^k) \\ Y^{k+1} = \nabla \operatorname{prox}_{\sigma f^*}(y^k + \sigma K \bar{x}^{k+1}) \cdot (Y^k + \sigma(K \bar{X}^{k+1} + \nabla_y \ell(x^k, y^k))), \end{cases}$$

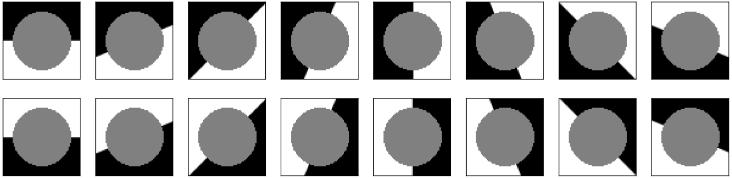
where we have used the fact that $\nabla \operatorname{prox}_{\tau g}(x) = (I + \tau D^2 g(\operatorname{prox}_{\tau g}(x)))^{-1}$.

Theorem ([Bogensperger, Chambolle, P. '22])

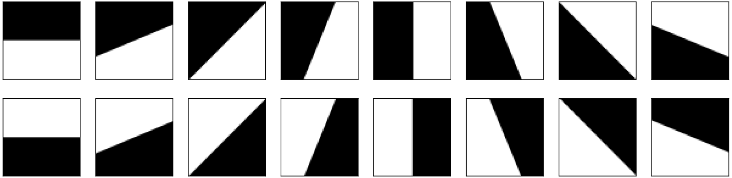
Assume that f^*, g are strongly convex and f, g^* are locally $C^{2,\alpha}$ then the piggyback primal-dual algorithm converges linearly.

Learning for inpainting

- We train on 64 images of size 64×64 with directions uniformly sampled between $[0, 2\pi]$ and we include random subpixel shifts.
- We train on a training set and evaluate on a test set.
- We experiment with different numbers of filters and different symmetry constraints for the filters.

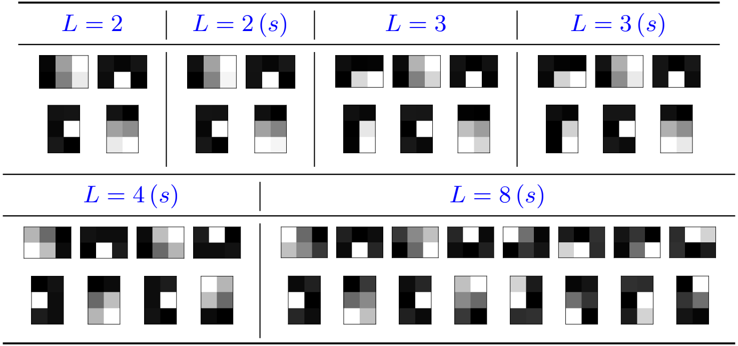


(a) Input images g_s



(b) Target images t_s

Results

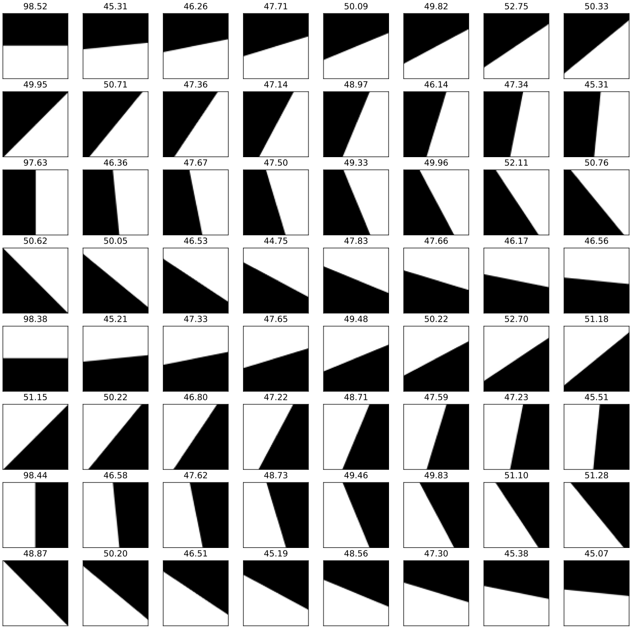


Data	FD	RT	CD	$L = 2$	$L = 2\,(s)$	$L = 3$	$L = 3\,(s)$	$L = 4\,(s)$	$L = 8\,(s)$
Train	135	195	6.69	1.26	1.22	1.19	1.27	0.85	0.77
Test	134	194	6.33	1.63	1.45	1.29	1.29	0.87	0.82

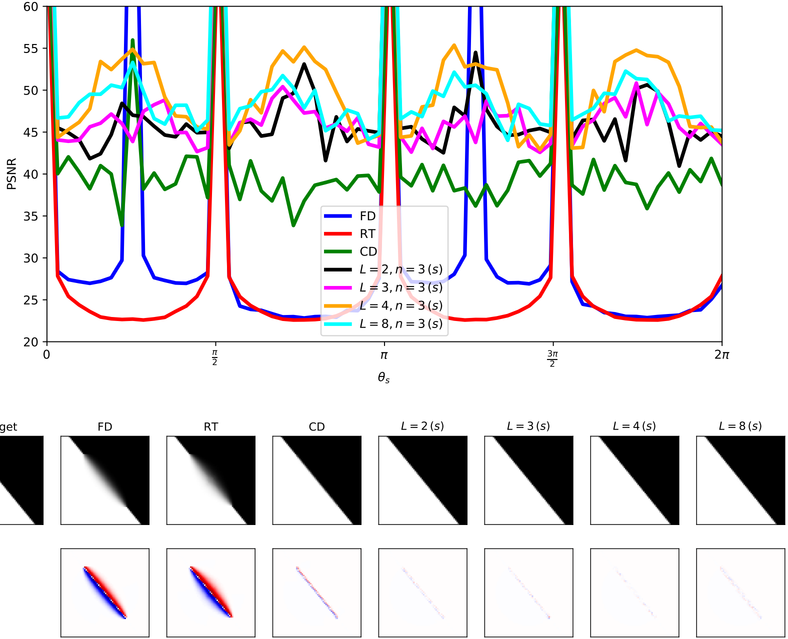
Table: $10^5 \times$ the mean squared error (MSE) of handcrafted and learned filters evaluated on both the training and test data.

Note that transpose symmetry is almost automatically learned!

Filter: $L = 8\,(s)$


98.52 45.31 46.26 47.71 50.09 49.82 52.75 50.33
49.95 50.71 47.36 47.14 48.97 46.14 47.34 45.31
97.63 46.36 47.67 47.50 49.33 49.96 52.11 50.76
50.62 50.05 46.53 44.75 47.83 47.66 46.17 46.56
98.38 45.21 47.33 47.65 49.48 50.22 52.70 51.18
51.15 50.22 46.80 47.22 48.71 47.59 47.23 45.51
98.44 46.58 47.62 48.73 49.46 49.83 51.10 51.28
48.87 50.20 46.51 45.19 48.56 47.30 45.38 45.07

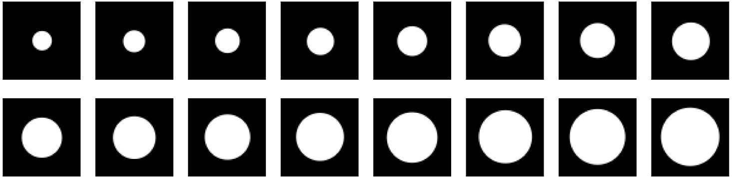
Comparison



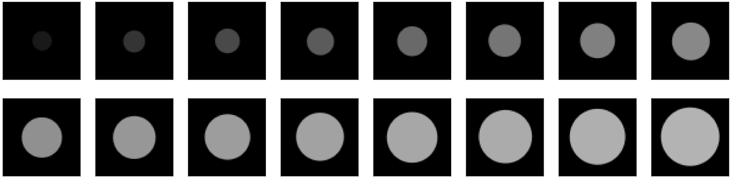
60
55
50
45
40
35
30
25
20
PSNR
0
$\frac{\pi}{2}$
π
$\frac{3\pi}{2}$
2π
θ_s
FD
RT
CD
$L = 2, n = 3\,(s)$
$L = 3, n = 3\,(s)$
$L = 4, n = 3\,(s)$
$L = 8, n = 3\,(s)$
Target
FD
RT
CD
$L = 2\,(s)$
$L = 3\,(s)$
$L = 4\,(s)$
$L = 8\,(s)$

Learning for disk regularization

- We train on 64 images with binary disks of various radii and subpixel shifted centers.
- The ground truth solutions can be computed with an explicit formula.
- We train on a training set and evaluate on a test set.
- We experiment with different numbers of filters and different symmetry constraints for the filters.

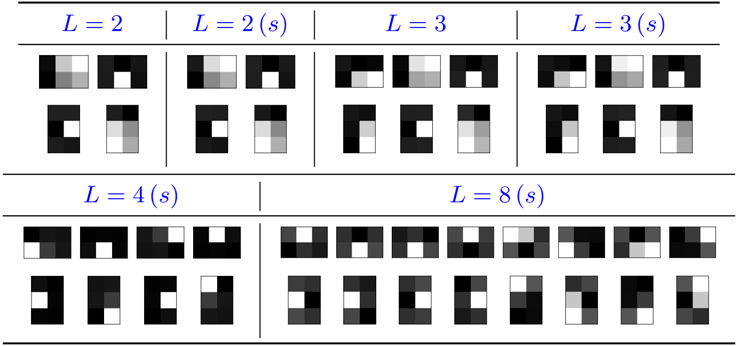


(a) Input images g_s



(b) Target images t_s

Results

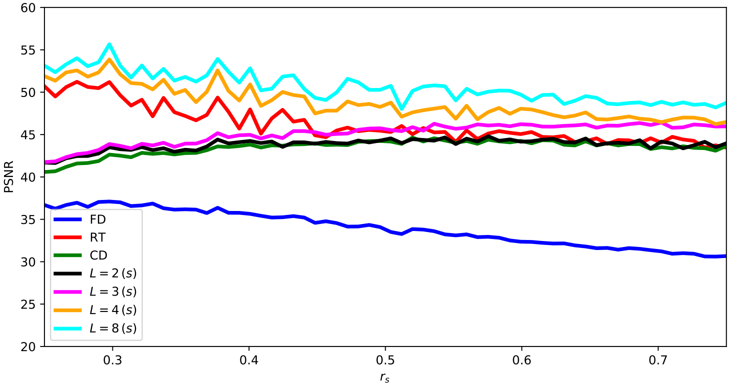


Data	FD	RT	CD	$L = 2$	$L = 2\,(s)$	$L = 3$	$L = 3\,(s)$	$L = 4\,(s)$	$L = 8\,(s)$
Train	22.28	1.36	2.33	2.10	2.10	1.62	1.63	0.73	0.48
Test	22.36	1.32	2.30	2.10	2.10	1.60	1.60	0.72	0.47

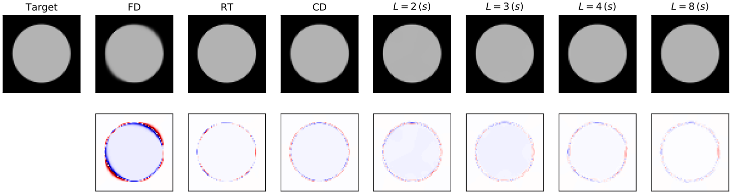
Table: 10^5 times the mean squared error (MSE) of handcrafted and learned filters for the disk denoising problem.

Observe that $L = 4\,(s)$ is very close to RT on cubic meshes!

Comparison



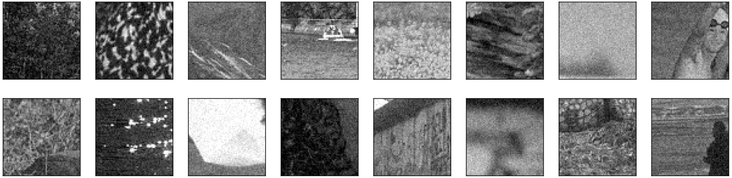
60
55
50
45
40
35
30
25
20
PSNR
0.3
0.4
0.5
0.6
0.7
r_s
FD
RT
CD
$L = 2\,(s)$
$L = 3\,(s)$
$L = 4\,(s)$
$L = 8\,(s)$



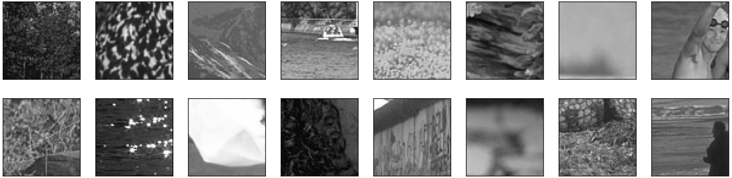
Target
FD
RT
CD
$L = 2\,(s)$
$L = 3\,(s)$
$L = 4\,(s)$
$L = 8\,(s)$

Natural image denoising

- We extract 64 patches of size 64×64 from a natural image database.
- The input images g_s contain 5% Gaussian noise.
- We learn both the filter weights and the regularization parameter λ by projecting on the set of filters with sum equals $\lambda > 0$.



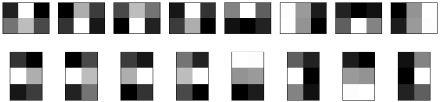
(a) Input images g_s



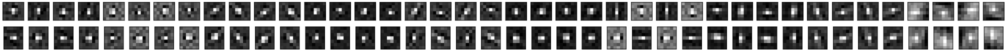
(b) Target images t_s

Results

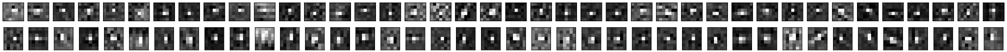
$L = 8\,(s),\ 2 \times 2$



$L = 40\,(s),\ 6 \times 6$



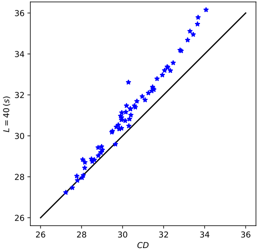
$L = 40,\ 6 \times 6$



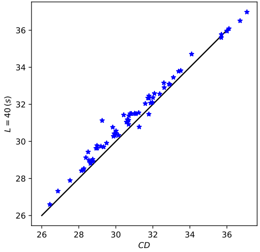
Data	FD	RT	CD	$L = 8\,(s)$	$L = 40\,(s)$	$L = 40$
Train	5.05	5.33	4.87	4.58	4.31	4.22
Test	4.72	5.05	4.51	4.28	4.10	4.13

Table: $10^4 \times$ the mean squared error (MSE) of handcrafted and learned filters for natural image denoising.

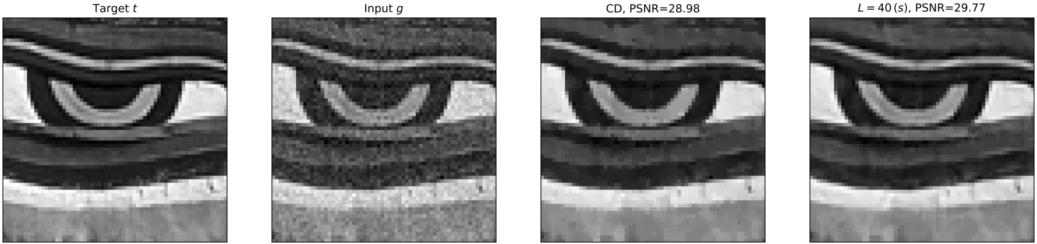
Comparison



(a) Training set



(b) Test set



(c) Example from the test set

Crossover experiments

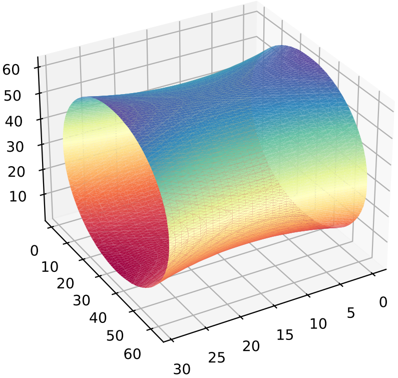
- How well do the learned filters generalize to other tasks?
- We compare the filters $L = 8\,(s)$ which gave good results on all tasks.

		Learning task			Handcrafted
		Line	Disk	Natural	CD
Evaluation task	Line	**0.82**	243.55	50.71	6.33
	Disk	1.88	**0.47**	4.08	2.30
	Natural	48.68	49.65	**42.80**	45.10

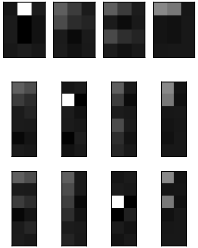
- The filters learned for inpainting generalize best, but there is no universal best discretization.

Extension to 3D

- Very recently, we have extended the learning of the discrete total variation to 3D
- We learn 4 sets of 3D filters on 3D minimal surface problems for which closed form solutions are available



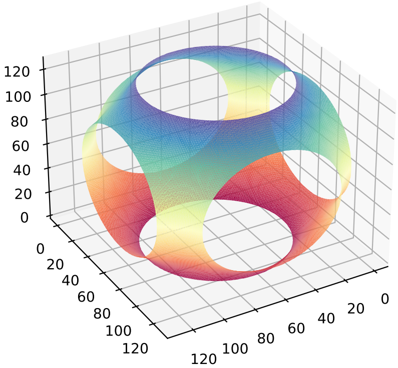
Catenoid



3D filters

Computing the Schwarz P surface

- After learning, we can compute high-accuracy minimal surfaces for which no closed for solution is available.
- A well-known example is the Schwarz P surface



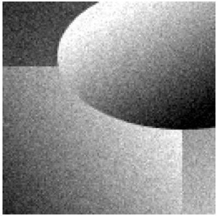
Total generalized variation

- In recent work [Bogensperger, Chambolle, Effland, P. '23] we have extended the framework to the secons order total generalized variation [Bredies, Kunsich, P. '10]

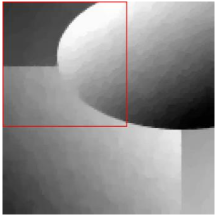
$$\mathsf{TGV}_\alpha^2(u) = \sup_p \left\{ \int_\Omega u \ \operatorname{div}^2 p \ \mathrm{d}x : p \in \mathcal{C}^\infty(\Omega, \mathsf{Sym}^{2\times 2}), \|p\|_\infty \leq \alpha_0, \|\operatorname{div} p\|_\infty \leq \alpha_1 \right\},$$

- In contrast to the total variation, the second order TGV can reconstruct piecewise affine images.
- In [Hosseini, Bredies '22] a Condat-like discretization was proposed, which served as the starting point for our work.

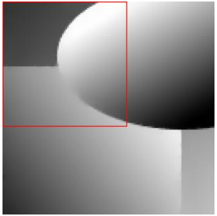
Example



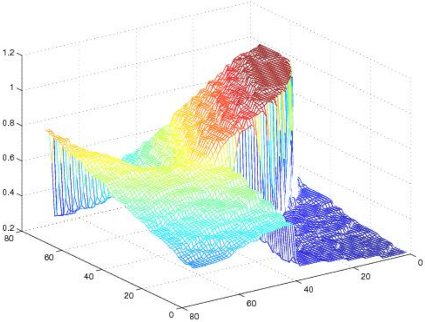
(a) Noisy



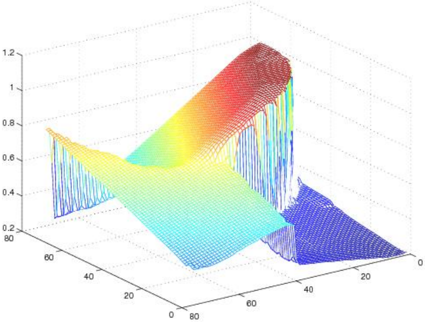
(b) TV



(c) TGV2



(d) Graph of TV



(e) Graph of TGV2

Discrete model

- The second-order TGV discretization in its primal form is given by

$$TGV(u) = \min_w \alpha_1 \|Du - w\| + \alpha_0 \|Ew\|,$$

where D is again the finite differences operator $Du = ((Du)^1, (Du)^2)$, where

$$\begin{aligned}
(Du)^1_{i+\frac{1}{2},j} &= \tfrac{1}{h}(u_{i+1,j} - u_{i,j}) && i \leq M-1, j \leq N,\\
(Du)^2_{i,j+\frac{1}{2}} &= \tfrac{1}{h}(u_{i,j+1} - u_{i,j}) && i \leq M, j \leq N-1.
\end{aligned}$$

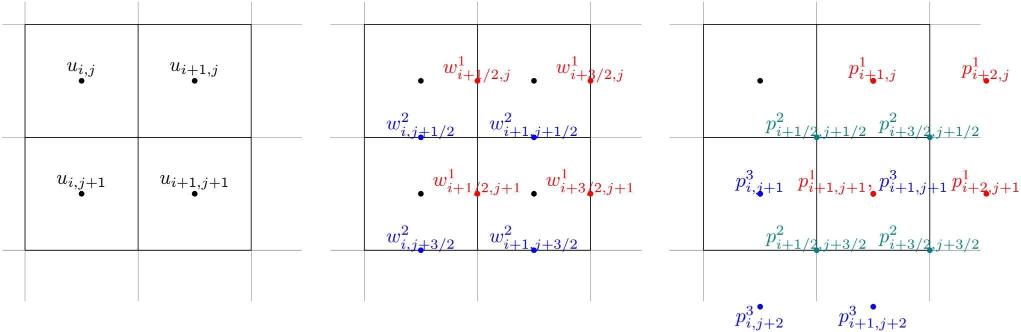
and E is the symmetrized vectorial gradient operator given by $Ew = \begin{pmatrix} (Ew)^1 & (Ew)^2 \\ (Ew)^2 & (Ew)^3 \end{pmatrix}$ with

$$\begin{aligned}
(Ew)^1_{i+1,j} &= \tfrac{1}{h}(w^1_{i+\frac{3}{2},j} - w^1_{i+\frac{1}{2},j}) && i \leq M-1, j \leq N,\\
(Ew)^2_{i+\frac{1}{2},j+\frac{1}{2}} &= \tfrac{1}{2h}(w^1_{i+\frac{1}{2},j+1} - w^1_{i+\frac{1}{2},j} + w^2_{i+1,j+\frac{1}{2}} - w^2_{i,j+\frac{1}{2}}) && i \leq M-1, j \leq N-1,\\
(Ew)^3_{i,j+1} &= \tfrac{1}{h}(w^2_{i,j+\frac{3}{2}} - w^2_{i,j+\frac{1}{2}}) && i \leq M, j \leq N-1.
\end{aligned}$$

- Now, introducing again interpolation operators L and K, applying some "convexity magic" the model becomes

$$TGV(u) = \sup_p \langle u, \operatorname{div}^2 p\rangle, \text{ s.t. } \|L \operatorname{div} p\|^*_Z \leq \alpha_1, \ \|Kp\|^*_Z \leq \alpha_0.$$

Pixel grid



$u_{i,j}$
$u_{i+1,j}$
$u_{i,j+1}$
$u_{i+1,j+1}$
$w^1_{i+1/2,j}$
$w^1_{i+3/2,j}$
$w^2_{i,j+1/2}$
$w^2_{i+1,j+1/2}$
$w^1_{i+1/2,j+1}$
$w^1_{i+3/2,j+1}$
$w^2_{i,j+3/2}$
$w^2_{i+1,j+3/2}$
$p^1_{i+1,j}$
$p^1_{i+2,j}$
$p^2_{i+1/2,j+1/2}$
$p^2_{i+3/2,j+1/2}$
$p^3_{i,j+1}$
$p^1_{i+1,j+1}$
$p^3_{i+1,j+1}$
$p^1_{i+2,j+1}$
$p^2_{i+1/2,j+3/2}$
$p^2_{i+3/2,j+3/2}$
$p^3_{i,j+2}$
$p^3_{i+1,j+2}$

The discrete TGV model is given by

$$\mathsf{TGV}^2_{\alpha,h}(u^h) = \min_{w^h, v^h_K, v^h_L} \left\{ h^2 \alpha_1 \|v^h_L\|_Z + h^2 \alpha_0 \|v^h_K\|_Z : L^*_h v^h_L = D_h u^h - w^h,\ K^*_h v^h_K = E_h w^h \right\}$$
$$= \sup_{p^h} \left\{ h^2 \langle \operatorname{div}^2_h p^h, u^h \rangle : \|L_h \operatorname{div}_h p^h\|^*_Z \leq \alpha_1, \|K_h p^h\|^*_Z \leq \alpha_0 \right\}.$$

Theorem

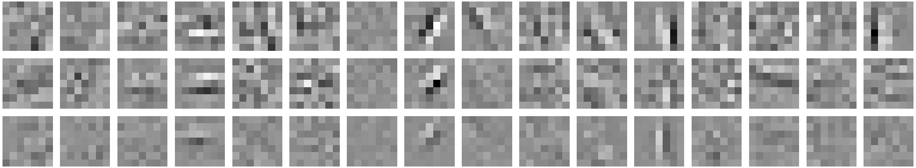
We consider the setting where u is affine plus periodic with period 1 in $\mathbb{R}^2$, and w is 1-periodic. Then, for interpolation operators K and L that have local support and bounded filter coefficients, $\mathsf{TGV}^2_{\alpha,h}(u^h)$ Γ-converges to $\mathsf{TGV}^2_{\alpha}(u)$.

Quantitative results for image denoising

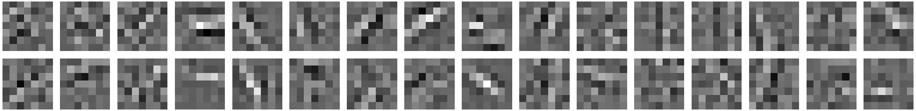
Table: Quantitative comparison of natural image denoising of the test set with 5% and 10% Gaussian noise for different handcrafted and learned discretizations.

	5% Gaussian noise			10% Gaussian noise		
	PSNR	MSE $\cdot 10^{-2}$	SSIM	PSNR	MSE $\cdot 10^{-2}$	SSIM
Corrupted f	26.04	0.2490	0.7885	20.02	0.9959	0.5382
TV	30.14	0.1049	0.9249	26.52	0.2445	0.8497
TGV	30.2	0.1043	0.9257	26.56	0.2431	0.8512
Handcrafted Disc. n_K=1, n_L=3	30.24	0.1046	0.9267	26.69	0.2394	0.8553
Handcrafted Disc. n_K=4, n_L=4	30.29	0.1030	0.9278	26.71	0.2370	0.8565
Learned Disc. n_K=1, n_L=3, 3×3	30.52	0.0935	0.9274	26.95	0.2172	0.8596
Learned Disc. n_K=4, n_L=4, 3×3	30.66	0.0906	0.9298	27.06	0.2123	0.8620
Learned Disc. n_K=8, n_L=8, 7×7	30.74	0.0896	0.9314	27.14	0.2090	0.8649
Learned Disc. n_K=8, n_L=8, 7×7, sym.	30.72	0.0898	0.9311	27.15	0.2089	0.8649
Learned Disc. n_K=10, n_L=10, 7×7	30.73	0.0896	0.9313	27.17	0.2081	0.8657
Learned Disc. n_K=16, n_L=16, 7×7	30.77	0.0891	0.9319	27.16	0.2087	0.8654
Learned Disc. n_K=16, n_L=16, 7×7, sym.	**30.77**	**0.0890**	**0.9320**	**27.18**	**0.2074**	**0.8659**

Learned filters



learned filters K with n_K=16



learned filters L with n_L=16

Figure: Learned 7×7 filters using n_L=16 and n_K=16 for denoising (10% Gaussian noise). The row of a depicted filter denotes the component of the respective vector/tensor field that it acts upon, whereas the column refers to the specific filter r or l (with $r = 1, \cdots, n_K$ and $l = 1, \cdots, n_L$.)

Example results

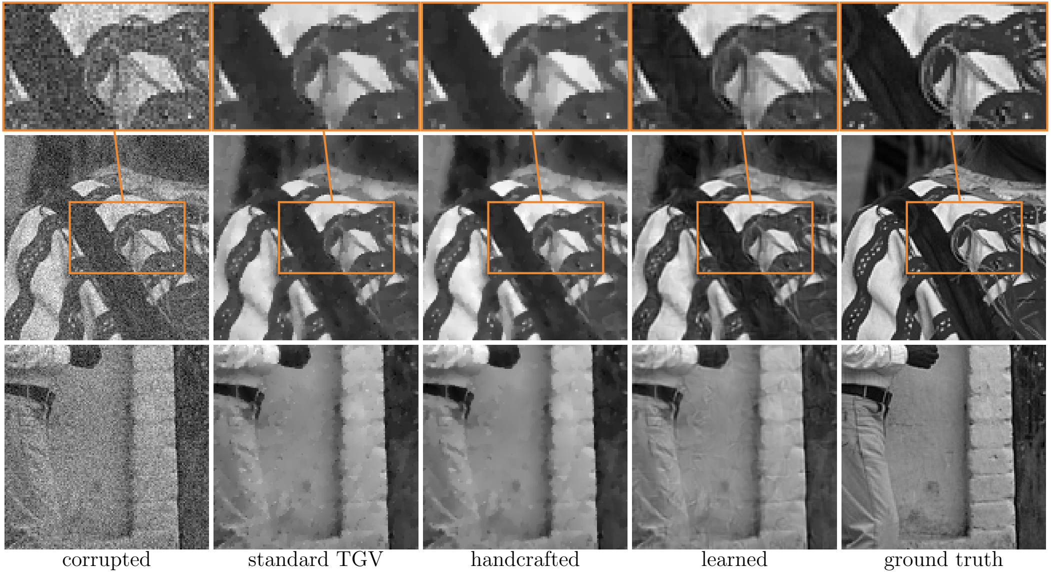


Figure: Sample reconstructions from natural test images (10% Gaussian noise) comparing the standard TGV, the handcrafted discretization scheme with n_K=1, n_L=3, and learned filters using n_K=16, n_L=16. For completeness, the ground truth images are also shown.

Beyond total variation regularization

- In [Bogensperger, Chambolle, P. '22], we applied the same learning framework to the shearlet transform [Kanghui, Kutyniok, Labate '06].
- A shearlet at scale j and shearing k is defined as.

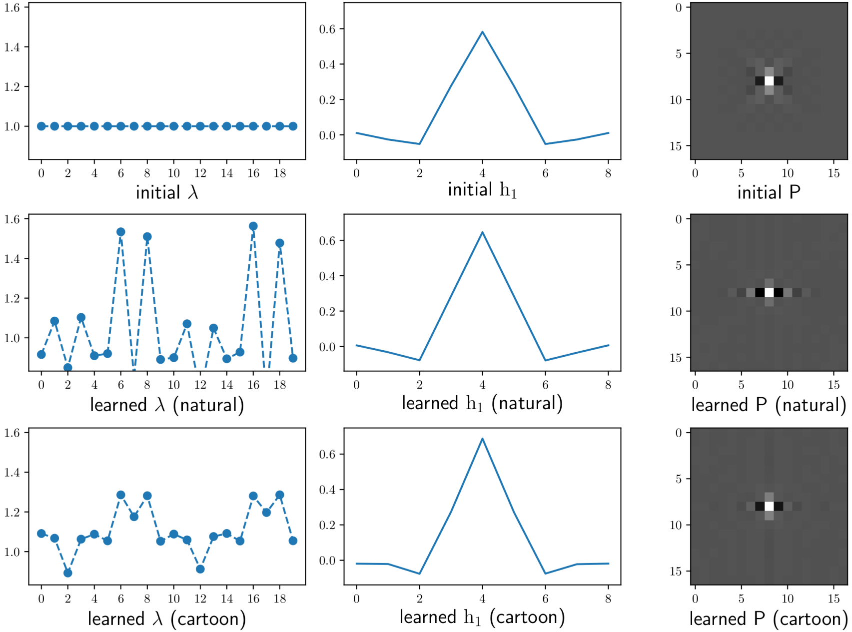
$$\psi^d_{j,k} = \left[\left(S_k\left((p_j * W_j)_{\uparrow 2^{j/2}} *_1 h_{j/2}\right)\right) *_1 \bar{h}_{j/2}\right]_{\downarrow 2^{j/2}},$$

  which essentially is constructed from a 1D low-pass filter h_1 and anisotropic 2D filter P. Additionally we also learn the importance weight $\lambda_{j,k}$ of each shearlet.
- Shearlets provide a multiscale framework similar to wavelets but better suited for encoding anisotropic features necessary for an efficient sparse representations of cartoon-like images.
- We tried to further optimize the shearlets using the piggy-back primal dual algorithm based on smooth-ℓ_1-regularized image denoising model

$$\min_u \|K(\theta)u\|_{1,\varepsilon} + \frac{1}{2}\|u - z\|_2^2,$$

  where θ is a placeholder for all learnable parameters and $\|\cdot\|_{1,\varepsilon}$ refers to a $C^{2,1}$ approximation of the ℓ_1 norm.
- Experiments are carried out on both natural images and synthetic piecewise affine images and we experimented with different settings of the smoothness parameter ε.

Learned shearlets



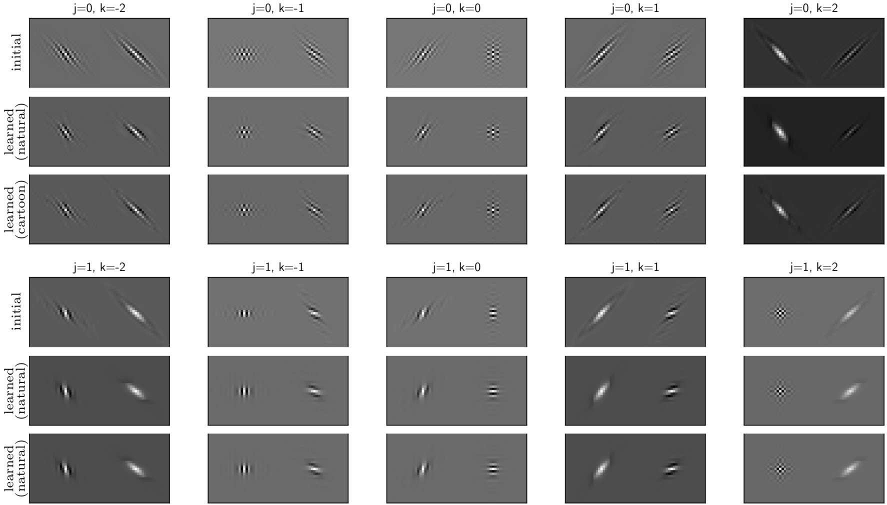
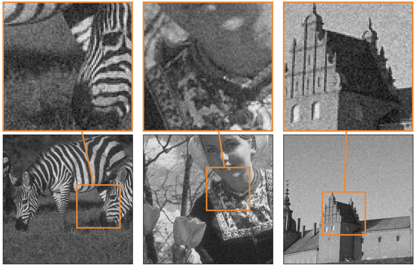
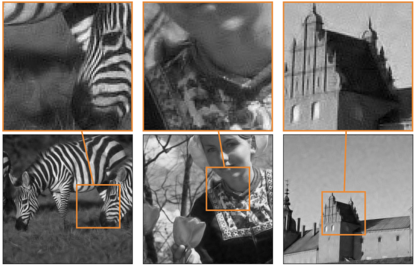

j=0, k=-2
j=0, k=-1
j=0, k=0
j=0, k=1
j=0, k=2
initial
learned (natural)
learned (cartoon)
j=1, k=-2
j=1, k=-1
j=1, k=0
j=1, k=1
j=1, k=2
initial
learned (natural)
learned (natural)

Image denoising

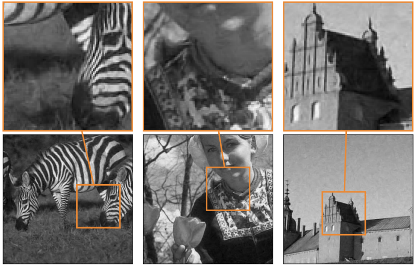
(a) ground truth sample test images t_l.



(b) noisy sample images z_l.



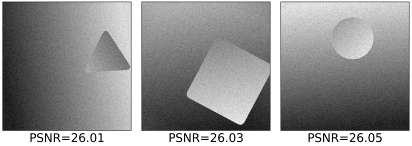
(c) sample denoised images u_l with initial shearlet system parameters and global $\lambda = 1.33$.



(d) sample denoised images u_l with learned shearlet system parameters shown in Figure 7.1.

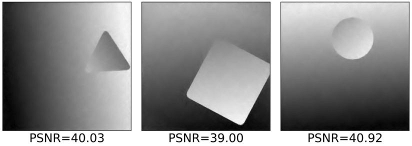
Image denoising

(a) ground truth sample test images t_l.



(b) noisy sample images z_l.

(c) sample denoised images u_l with initial shearlet system parameters and global $\lambda = 2.04$.



(d) sample denoised images u_l with learned shearlet system parameters shown in Figure 7.1.

Quantitative evaluation

Comparison between the performance of the initial shearlet transform (hand tuned, $\lambda = 1.33$ and $\lambda = 2.04$, resp.) and the optimized shearlet transform for various setting of the smoothing parameter ε on the natural (N) and cartoon-like (C) dataset. Note that the learned transform clearly outperforms the initial shearlet transform and that smaller settings of ε lead to better results.

		$\varepsilon = 10^{-1}$		$\varepsilon = 10^{-2}$		$\varepsilon = 10^{-3}$		$\varepsilon = 10^{-4}$		$\varepsilon = 0$	
		MSE	PSNR	MSE	PSNR	MSE	PSNR	MSE	PSNR	MSE	PSNR
N	Initial	0.001929	27.15	0.001187	29.38	0.001056	30.05	0.00105	30.1	0.00105	30.1
	Optimized	0.001148	29.56	0.000931	30.53	0.000821	31.14	0.000813	31.2	0.000813	31.2
C	Initial	0.001392	28.57	0.000345	34.62	0.000155	38.14	0.000134	38.79	0.000132	38.84
	Optimized	0.000293	35.44	0.000165	37.87	0.000115	39.45	0.000103	39.92	0.000101	40.02

Conclusion

- We proposed learning optimized finite differences discretizations of the total variation.
- The learning is constraint to a class of consistent discretizations which Γ-converge to the continuous total variation.
- We proposed a piggy-back primal-dual algorithm for computing derivatives.
- Symmetry constraints on the filters give better generalizations.
- The learned discretizations give significant improvements when optimized for certain applications but no best universal discretization could be learned.
- The learning framework has been extended to 3D TV and more complex regularization operators such as TGV and shearlets.

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Thank you for your attention!